INCH-POUND

MIL-R-12323G
31 December 1992
SUPERSEDING
MIL-R-12323F
9 June 1975

MILITARY SPECIFICATION

REFRIGERATORS AND RELATED EQUIPMENT, PACKAGING AND PACKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers optional detailed requirements or performance requirements for packaging, packing, and marking of mechanically cooled refrigerators, freezers, and related equipment with storage volume up to 100 cubic feet.

1.2 Classification.

1.2.1 Options. Packaging and packing requirements shall be of the following options as specified (see 6.2).

Option I - Detailed requirements
Option II - Performance requirements

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
A-A-55057	-	Panels, Wood/Wood; Construction and Decorative
L-P-378	-	Plastic, Sheet and Strip, Thin Gauge, Polyolefin
TT-P-664	-	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, Voc-Compliant
UU-P-134	-	Paper, Wrapping, Wet-Waxed
UU-T-81	-	Tags, Shipping and Stock
PPP-B-576	-	Boxes, Wood-Cleated Panelboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping Fiberboard
PPP-B-640		Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible
PPP-P-291	-	Paperboard, Wrapping and Cushioning
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packaging Of
PPP-T-60	-	Tape: Packaging, Waterproof
PPP-T-97	-	Tape: Packaging/Industrial, Filament Reinforced

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation Methods Of
MIL-C-11133	-	Crates, Shipping, Wood, Open, Wirebound
MIL-C-52950	-	Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-1153	-	Refrigerators and Related Equipment, Packaging and Packing of: Container, Cleated-Plywood, and Container Cleated Fiberboard, for Levels A and B Shipments
5-13-1172	-	Refrigerators and Related Equipment, Packaging and Packing of: Method of Securing Units Within Exterior Shipping Containers

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3950	-	Standard Specification for Strapping, Plastic (and Seals)
D 3951	-	Standard Practice for Commercial Packaging
D 3953	-	Standard Specification for Strapping, Flat Steel and Seals
D 4727	-	Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes

(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document takes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

3.1.1 Level A. Unless otherwise specified, units shall be preserved and packaged either in accordance with option I or II. When specified (see 6.2), units shall be preserved and packaged in accordance with option I.

3.1.1.1 Option I - detailed requirements.

3.1.1.1.1 Disassembly and matchmarking.

3.1.1.1.1.1 Disassembly. All detachable parts and accessories, such as trays, shelves, food jars, and other parts liable to be damaged or cause damage during handling and shipment, shall be removed from the unit and packaged as specified in 3.1.1.1.12. Bolts, nuts, screws, washers, pins, and other connectors removed during disassembly shall be placed in a cotton cloth drawstring bag and tied to the unit or disassembled part, or packaged in the same fiberboard box as the accessories and parts. Alternatively, detached parts and accessories for units packed as specified in option II, may be secured in their normal position in a manner to prevent damage.

3.1.1.1.2 Matchmarking. When detached parts and accessories may cause difficulty in reassembly, they shall be matchmarked. Matchmarking information shall be on shipping tags conforming to type A of UU-T-81, attached to mating parts.

3.1.1.1.3 Draining. All water-containing parts, such as automatic ice cube makers, water cooling systems, and water-cooled condensers, shall be completely drained to prevent damage from freezing.

3.1.1.1.4 Cleaning. Cleaning, application and type of preservative, and method of preservation shall be in accordance with MIL-P-116 as specified herein. All exposed, uncoated, ferrous metal surfaces of the unit, detached parts, and accessories shall be thoroughly cleaned in accordance with process C-1 and thoroughly dried.

3.1.1.1.4.1 Application of preservative. All exposed, uncoated, ferrous metal surfaces, except corrosion-resisting steel, within the food storage or freezer compartments, shall be coated with type P-14 preservative. All other exposed uncoated, ferrous metal surfaces, except corrosion-resisting steel, shall be coated with type P-1 preservative. Pulleys and shafts shall be coated as specified in 3.1.1.1.8.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the current version before use.

3.1.1.1.4.2 Protection of preservative. Surfaces coated with type P-14 preservative from which preservative might be removed by contact with packaging and packing materials, shall be covered, wrapped, or bagged with greaseproof barrier material in accordance with method I.

3.1.1.1.5 Sealing. All exposed ends or wires, plug openings, and sockets shall be sealed with tape conforming to type III, class 1 of PPP-T-60.

3.1.1.1.6 Electrical components. Detached electrical components, such as motors, generators, control boxes, and temperature and pressure control switches shall be preserved by method II, with selection of the applicable submethod. Attached electrical components not inclosed within the storage compartment of the refrigerator shall have all openings into their interiors, such as inlets, outlets, vents, valves, and louvers sealed with tape as specified in 3.1.1.1.5.

3.1.1.1.7 Operating instruction. Operating instructions and service manuals shall be individually placed within a commercial paper envelope and secured within the interior of the unit with pressure-sensitive tape as specified in 3.1.1.1.5.

3.1.1.1.8 Servicing of refrigerating unit. The refrigerating system shall be serviced for shipment according to the manufacturer's recommended procedure and as specified herein. Drive belts shall be removed, and the exposed, uncoated, ferrous metal surfaces of pulleys and shafts cleaned in accordance with process C-1 and coated with primer conforming to TT-P-664. When primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. All floating mechanisms and tubing not rigidly fixed, such as vibration loops, free coils, capillary tube loops and coils, shall be secured in a fixed position to minimize vibration and subsequent damage during handling and shipment. Floating mechanism is defined as any part that is inherently flexible or movable, or otherwise mounted which might vibrate during shipment.

3.1.1.1.9 Doors, drawers, and bins. Interior doors, drawers, and bins not covered in 3.1.1.1.1.1 shall be secured in the closed position in a manner to prevent movement or damage while in transit. Exterior doors, drawers, and bins of the unit shall be secured by one of the following methods:

(a) Locking. When means of locking is provided, the doors, drawers, and bins shall be closed, latched, and locked in a secure position. When locking is by means of a padlock, the doors and bins shall be closed and locked by means of a bolt and nut in a manner to prevent them from opening. When doors, drawers, and bins are provided with built-in locks they shall be closed, locked, and the keys placed in a cotton cloth drawstring bag and the bag secured to one of the handles. The bagged keys and handle shall then be wrapped with tape specified in 3.1.1.1.5.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the current version before use.

(b) Strapping. When means of locking doors, drawers, and bins are not provided, they shall be closed and secured with steel strapping conforming to class 1, type I of ASTM D 3953 or nonmetallic strapping conforming to type II or III of ASTM D 3950. Steel strapping shall be minimum 1/4 by 0.015 inch on household type refrigerators and minimum 3/8 by 0.020 inch on all other type units positioned above or near the latches. The use and positioning shall be the same for nonmetallic strapping. A continuous strip of cushioning material minimum 1-1/2 inches wide and 1/2 inch thick, conforming to grade III, class B, style optional size F of A-A-1898, shall be positioned under the steel strapping. The nonmetallic strapping shall be minimum 3/8 inch by 0.015 inch on all other type units positioned above or near the latches. The strapping shall be centered on the cushioning material and shall completely encircle the unit in a horizontal position to prevent contact of the steel strapping with the finished surfaces of the unit. The clips or buckles on nonmetallic strapping shall not bear on the finished surfaces of the front, sides, or top of the unit. Tensioning of strapping shall not mar or distort the surface of the unit.

(c) Taping. Alternative to strapping, the doors, drawers, or bins shall be secured in a closed position with 1 inch minimum width tape conforming to type III of PPP-T-97 extending around the entire unit and overlapping itself not less than six inches, or by not less than four strips of tape, each a minimum of 18 inches in length, applied in an "X" position to the latch or unhinged side of the door and side. The tape shall show no staining when tested in accordance with 4.3.1.

3.1.1.1.10 Legs. Detachable legs shall be removed from the unit and, together with attaching hardware, shall be cushioned and packaged as specified in 3.1.1.1.12. Leveling pads on detachable legs shall be completely screwed into the legs. Permanently attached legs shall have the leveling pads removed (unless pads are permanently assembled) and packaged with accessories and parts.

3.1.1.1.11 Line cord. The nondetachable line cord shall be looped and securely tied in place at the relay bracket, or similar location, or to the back of the unit. A voltage and frequency notice shall be secured to the cord, except when such information appears on the manufacturer's nameplate.

3.1.1.1.12 Accessories and parts. Glass shelves, trays, food jars, covers, and similar fragile items, metal or wire shelves, accessories, and parts shall be individually wrapped with cushioning material conforming to grade III, class B, any style or size of A-A-1898 or PPP-P-291 and packaged in a fiberboard box or boxes conforming to grade V3c, V3s, or V4s, style optional, of PPP-B-636. Each box shall be closed in accordance with method III as specified in the appendix of PPP-B-636. When total gross weight of the packaged accessories and parts is 65 pounds or less, the fiberboard box or boxes shall be secured or blocked and braced within the unit in a manner to prevent movement during handling and shipment. When adequate means are

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

not available to secure the fiberboard box within the unit, or when total gross weight exceeds 65 pounds, the accessories and parts shall be packed as specified in 3.3.

3.1.1.1.13 Tags. A tag or tags conforming to type A of UU-T-81 shall be attached in a conspicuous place on each unit and give detailed instructions for deprocessing or placing the unit into service.

3.1.1.2 Option II - performance requirements.

3.1.1.2.1 Preservation and packaging. Preservation and packaging shall be as specified in 3.1.1.1 through 3.1.1.1.8, 3.1.1.1.10, 3.1.1.1.11, and 3.1.1.1.13.

3.1.1.2.2 Doors, drawers, and bins. Doors, drawers, and bins shall be packaged as specified in 3.1.1.1.9 or in a manner to prevent movement or damage to the component or unit during shipment or handling. As a minimum, there shall be no evidence of damage to these components when the packed unit is tested for the applicable level of pack in PPP-P-600.

3.1.1.2.3 Accessories and parts. Accessories and parts shall be packaged as specified in 3.1.1.1.12, or they may be secured in their normal position or relocated within the unit and secured to prevent damage to the part or unit. Accessories and parts, secured in place or relocated within the unit, shall not be damaged or cause damage to the unit when tested as specified for the applicable level of pack of PPP-P-600.

3.1.2 Level C. The units shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets the requirements of ASTM D 3951.

3.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

3.2.1 Level A. Unless otherwise specified units shall be packed either in accordance with option I or II. When specified (see 6.2), units shall be packed in accordance with option I.

3.2.1.1 Option I - detailed requirements. Each unit, together with accessories and parts, preserved and packaged as specified in 3.1.1.1.4 or 3.1.1.1.4.1, shall be packed in a shipping container as specified in table I.

3.2.1.1.1 Boxes, wood-cleated, panelboard - maximum weight of contents 200 pounds. Wood-cleated, fiberboard and veneer-paper overlaid boxes shall conform to type I or III, class 2, style A or B of PPP-B-576, except as shown on Drawing 5-13-1153. Each shipping container shall be strapped as shown on Drawing 5-13-1153.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

3.2.1.1.2 Boxes, wood, cleated-plywood - maximum weight of contents 1,000 pounds. Wood, cleated-plywood boxes shall conform to overseas type, style A or B, grade A or B, type 2 load of PPP-B-601, except as shown on Drawing 5-13-1153. Each shipping container shall be strapped as shown on Drawing 5-13-1153.

3.2.1.1.3 Boxes, wood, nailed and lock corner - maximum weight of contents 1,000 pounds. Wood boxes shall conform to class 2, style 7, grade A or B, type 3 load of PPP-B-621, except that reinforcing members shall be of sufficient width to ensure complete contact with fiberboard edge pads when refrigerators are secured method A or B (see table I). The top, sides, and ends of the container shall be lined with waterproof barrier material conforming to PPP-B-1055, with all joints have a 4 inch minimum single lap.

3.2.1.1.4 Crates, wood; lumber and plywood sheathed, nailed and bolted, weight of contents exceeding 500 pounds. Sheathed nailed wood crates shall conform to type I or II, class 1 or 2, style A of MIL-C-104, except for weight of contents up to and including 1,000 pounds the following shall apply:

(a) All structural members of the end panels shall be minimum 1 by 4 inch (nominal) lumber.

(b) All structural members of the side panels, except the vertical end struts, shall be nominal 1 by 4 inch lumber. The vertical end struts shall be nominal 2 by 4 inch lumber.

(c) A continuous sheet of minimum 1/2 inch thick plywood may be substituted for the 1/4 inch thick plywood, roofing felt, and wood sheathing on the top of narrow top plywood sheathed crates. The plywood shall conform to either type A or oriented strand board with phenalic exterior glue of type B of A-A-55057.

(d) The bolts used to prevent splitting of the 2 by 4 inch skids shall be minimum 1/4 inch diameter or maximum 3/8 inch diameter.

3.2.1.2 Option II - performance requirements. Each unit, preserved and packaged as specified in 3.1.1.2 or 3.1.2 shall be packed in accordance with the level A requirements of PPP-P-600. The packed unit shall meet the level A requirements of PPP-P-600.

3.2.2 Level B. Unless otherwise specified units shall be packed either in accordance with option I or II. When specified (see 6.2), units shall be packed in accordance with option I.

3.2.2.1 Option I - detailed requirements. Each unit, together with accessories and parts, preserved and packaged as specified in 3.1.1 or 3.1.2, shall be packed in a shipping container as specified in table I. The contents of each open type crate shall be protected from the elements by a full length shroud fabricated from minimum 4 mil thick polyethylene

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

conforming to L-P-378 and the seams shall be heat sealed. The shroud shall hang free, when possible, and shall extend to the base of the crate. All sharp points of contact between the item, the blocking and bracing, and the shroud shall be cushioned to prevent rupture or chafing of the shroud. All open type crates shall be assembled, closed, and strapped in accordance with the appendix of the applicable crate specification.

3.2.2.1.1 Boxes, shipping, fiberboard. The fiberboard shipping containers shall be snug-fitting conforming to style RSC, type CF or SF, class domestic of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells. Each fiberboard shipping container shall be securely closed in accordance with method II as specified in the appendix of the container specification. When specified (see 6.2), the fiberboard shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

3.2.2.1.2 Boxes, fiberboard, corrugated, triple-wall - maximum weight of contents 275 pounds. Triple-wall, corrugated, fiberboard boxes shall conform to class 1, grade A, style E of PPP-B-640. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of material specified in 3.2.2.1.1. Each fiberboard shipping container shall be securely closed in accordance with the appendix of the container specification. When specified (see 6.2), the triple-wall corrugated shipping containers shall be fabricated of class 2, weather-resistant material.

3.2.2.1.3 Boxes, wood-cleated panelboard - maximum weight of contents 400 pounds. Wood-cleated, panelboard containers shall conform to type I or III, class 1 of PPP-B-576 and as shown on Drawing 5-13-1153. When specified (see 6.2), class 2 weather-resistant material shall be used in the fabrication of the shipping container.

3.2.2.1.4 Boxes, wood, cleated-plywood - maximum weight of contents 1,000 pounds. Wood, cleated-plywood shipping containers shall conform to domestic type, grade B of PPP-B-601 and as shown on Drawing 5-13-1153.

3.2.2.1.5 Crates, shipping, wood, open, wirebound - maximum weight of contents 1,000 pounds. Wirebound shipping crates shall conform to type II, class 1 or 2, style 1 or 2, grade B or type III, class 1, 2, or 3, style 1 or 2, grade B of MIL-C-11133, as applicable.

3.2.2.1.6 Crates, wood open and covered - maximum weight of contents 2,500 pounds. Open wood crates shall conform to style A, type I, II, IV, or V, class 1 of MIL-C-52950, as applicable.

3.2.2.1.7 Crates, wood, lumber and plywood sheathed, nailed and bolted - weight of contents exceeding 500 pounds. Sheathed, nailed, wood crates shall be as specified in 3.2.1.1.4, except that the waterproof barrier material shall not be required.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the current version before use.

3.2.2.2 Option II - performance requirements. Each unit, preserved and packaged as specified in 3.1.1.2 or 3.1.2, shall be packed in accordance with the level B requirements of PPP-P-600. The packed unit shall meet the level B performance requirements of PPP-P-600.

3.2.3 Level C. Units, preserved and packaged as specified in 3.1, shall be secured and packed in shipping containers in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers and packing shall be in accordance with ASTM D 3951, or meet the level C requirements of PPP-P-600.

3.3 Securing of units in the shipping container - option I and II. Each complete unit shall be secured within the shipping container by the methods as indicated in table I for option I for the weight limitations specified, except for level C shipments. Each complete unit shall be secured in a manner to prevent movement or damage for option II. A minimum of 1 inch clearance shall be provided between the top, sides, and ends of the unit and the adjacent inside surfaces of the shipping container. The fiberboard box or boxes containing accessories and parts not secured within the unit shall be secured in an unused portion of the same shipping container in a manner to prevent movement or damage to the accessories, parts, or unit during transit or storage.

TABLE I. Methods of securing units in shipping containers and maximum weight of contents for option I - Level A and B

Container	Securing method	Weight of Contents (pounds)
Boxes, shipping, fiberboard, PPP-B-636 1/	See 3.2.2.1.1	Weight of content shall not exceed the weight limitation of the specification
Boxes, fiberboard, corrugated 1/ triple-wall, PPP-B-640	See 3.2.2.1.2	275 Maximum
Boxes, wood-cleated panelboard PPP-B-576 and Drawing No. 5-13-1153	B 3/ A or C 3/ or A and C 3/	300 Maximum 2/ 400 Maximum 2/
Boxes, wood, cleated-plywood, PPP-B-601 and Drawing No. 5-13-1153	B 3/ A or C 3/ or A and C 3/	300 Maximum 700 Maximum
Boxes, wood, nailed and locked corner - PPP-B-621	B or C	800 Maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

TABLE I. Methods of securing units in shipping containers and maximum weight of contents for option I - Level A and B (cont'd)

Container	Securing method	Weight of Contents (pounds)
Crates, wood; lumber and plywood sheathed, nailed and bolted, MIL-C-104	A or C 3/ or A and C 3/	Exceeding 500
Crates, shipping, wood, open, wirebound 1/ MIL-C-11133	B A or C or A and C	300 Maximum 1,000 Maximum
Crates, wood, open and 1/ covered, MIL-C-52950	B A or C or A and C	300 Maximum 2,500 Maximum

1/ For level B shipments only.

2/ 200 maximum for level A shipment

3/ Cushioning material shall be water-resistant for level A shipment.

3.3.1 Method A (bolting). The unit shall be bolted to the base and one panel of the container and shall be cushioned over the full length of the two front-side edges of the unit. A minimum 1 inch clearance shall be provided between the outermost projections of the front, sides, and top of the unit and the adjacent inside surfaces of the container (see Drawing 5-13-1172).

3.3.1.1 Base. The unit shall be bolted to the base of the shipping container with carriage bolts extending through the load-bearing members or through the skids and load-bearing members, or through the skids of the base, and up through the base or legs of the unit. The heads of the carriage bolts shall be at the underside of the floorboard, load-bearing members, or the skids. Not less than four bolts shall be used for securing the unit to the crate base. Bolt diameter shall be not less than 3/8 inch for units weighing up to 500 pounds. Units in excess of 500 pounds shall require 1/2 inch bolts. Alternative to carriage bolts, units weighing less than 400 pounds may be secured by minimum 5/16 inch diameter machine bolts. Round flat washers and lockwashers shall be used under the heads of all machine bolts, and lockwashers shall be used with all nuts on carriage bolts.

3.3.1.2 Panel. The unit shall be secured to one panel of the container with two bolts, each bolt extending through a cleat on the outside face of the container, through the container panel, and through a back bar of group III or IV woods of MIL-STD-731, minimum 1-1/8 inches thick by 5-1/2 inches wide extending the full inside width of the container. Each bolt shall also extend through one of two blocking members of group III or IV woods, minimum 1-1/8 inches thick by 5-1/2 inches wide, blocking members extending to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

inside surface of the top and side panels of the container, with bolts extending into gusset plates which form a part of the rear face of the unit. The extending of the bolts through a cleat on the outside surface of the container may be omitted provided that two bracing members, minimum 3/4 inch thick by 1-3/4 inches wide, are secured to the inside surface of the top or side panels of the container and abut the blocking members bolted to the rear face of the unit.

3.3.1.3 Edge pads. Edge pads shall be made of built-up corrugated fiberboard. Refrigerators with top mounted units shall be provided with full length pads extending down from the top edge of the sides and front a minimum of 3 inches and of sufficient thickness to provide a tight pack and a minimum clearance of 1 inch between the outermost projections of the front and sides of the unit and the interior surface of the shipping container. All other units shall be provided with top edge and front and side edge pads. These pads shall be full length of the sides and front and shall extend in both directions (over the top and down the front and sides) a minimum of 3 inches. These pads shall be of the sufficient thickness to provide a tight pack and a minimum clearance of 1 inch between the top, front, and sides of the unit and the interior surface of the shipping container. The pads shall be secured with tape conforming to PPP-T-97.

3.3.2 Method B (pressure pack). The unit shall be cushioned and held securely within the shipping container by built up, corrugated, fiberboard, edge pads. Edge pads shall be provided on all horizontal edges and on all vertical edges of the unit. Edge pads shall be of sufficient width to extend 3 inches in from the edge onto top, bottom, sides and ends of the unit, and of sufficient thickness to provide a tight pack and a minimum 1 inch clearance between any protruding part of the unit and the adjacent inside surfaces of the container.

3.3.3 Method C (blocking and bracing). A double-faced, corrugated, fiberboard tray, minimum 4 inches deep, shall be placed on the top and bottom of the refrigerator. The tray shall fit the top and bottom of the refrigerator in a compact manner. Blocking and bracing shall be in accordance with the appendix of MIL-C-104, and as specified herein. A perimeter wood frame of sufficient width and thickness to prevent side or end movement of the unit, shall be nailed or bolted to the base of the container. A perimeter frame shall also be secured to the underside of the top frame members to eliminate top sway. When possible, the unit shall be bolted to the base of the container with a minimum of four 1/2 inch bolts as specified in 3.3.1.1. A minimum of 1 inch clearance shall be provided between the sides and ends of the unit, or any protruding part in front of the unit, and the interior surfaces of the shipping container.

3.3.3.1 Alternative to method C. Each unit, with top mounted refrigeration units, shall be anchored in accordance with the appendix of MIL-C-104 or each unit with top mounted refrigeration units shall be bolted to the base of the crate with not less than four 1/2 inch diameter carriage

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the current version before use.

bolts as specified in 3.3.1.1, and the base blocked with a perimeter wood frame of sufficient width and thickness to prevent side or end movement of the unit nailed to the base of the crate. Double-faced, corrugated, fiberboard, edge pads, having minimum width of 3 inches on each face, shall be placed between the bottom and finished surfaces of the sides and ends of the unit and wood frame and the base of the crate, and around the top edge of the cabinet extending down the sides and ends of the cabinet and be secured with tape conforming to PPP-T-97. The bottom fiberboard pads may be omitted when the perimeter wood frame contacts the structural steel base and not the finished surfaces of the sides and ends of the cabinet. A minimum clearance of 1 inch shall be maintained between any protruding part of the top, sides, and ends of the unit and the interior surfaces of the shipping container.

3.3.4 Protection of finished surfaces. One of the following shall be placed between the corrugated fiberboard or blocking members and the finished surfaces of the unit:

(a) A sheet of wax paper conforming to type III of UU-P-134.
(b) Cellulosic cushioning material conforming to grade I or II, class B of A-A-1898.

The face of cushioning material in direct contact with fiberboard pads or blocking members shall be surfaced with kraft paper having a minimum basis weight of 30 pounds per ream (24 by 36 - 500).

3.4 Marking.

3.4.1 Interior packages. Each package, box, and overwrap shall be marked or labeled to indicate the item and quantity contained therein. Component parts that have been overwrapped and secured in their respective locations shall not require marking.

3.4.2 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-192. Set marking requirements shall apply when accessories and parts are packed separate from the unit.

3.4.2.1 Precautionary marking. Special handling marking requirements in MIL-STD-129 applicable to fragile items, such as arrows, the words "UP" and "THIS SIDE UP", and CENTER OF BALANCE" shall apply.

3.4.2.2 Additional marking. The panel of the sheathed nailed wood crate, cleated fiberboard, cleated-plywood, or nailed wood shipping container against the front of the refrigerator being packed shall be marked "FRONT" in letters not less than 1 inch in height.

3.5 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the currentversion before use.

3.6 Workmanship. The end item shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The end item shall be examined for visual preparation for packaging, preservation and packaging, packing, and marking in accordance with the defects listed in tables II through V. The lot size shall be expressed in units of packaged or packed refrigerators or related equipment, as applicable. The sample unit shall be one packaged or packed refrigerator or related equipment, of the specified type, design and

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the currentversion before use.

characteristic. The inspection levels and the acceptable quality levels (AQLs), expressed in terms of defects per hundred units, shall be as follows:

Table No.	Inspection levels	AQLs Major	AQLs Total 1/
II	S-2	2.5	10
III	S-2	2.5	10
IV	S-2	2.5	10
V	S-2		6.5

1/ Total represents major and minor defects combined for the AQL value.

TABLE II. End item visual preparation for packaging defects

Examine	Defect	Category Major	Minor
Disassembly	Detachable parts and accessories not removed to prevent damage (when specified)	101	
	Detachable parts and accessories not secured in position (when specified)	102	
Matchmarking	Tags with information not attached to mating parts (when specified)		201
Draining	All water not removed from water containing parts of refrigerator	103	
Deprocessing tag	Unit not tagged to show deprocessing requirements		202

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the current version before use.

TABLE III. End item preservation and packaging defects

Examine	Defect	Category Major	Category Minor
Level A			
Option I			
Sealing exposed ends of wire, plug, openings, sockets, faces of gages, and instruction plates	Not sealed with tape when specified		201
Electrical components	Components not preserved as specified	101	
	Openings into interiors not taped		202
Operating instructions	Not placed individually in envelope		203
Refrigerating units	Drive belts not mounted in place with tension relaxed		204
Floating mechanism and tubing	Not secured in fixed position	102	
Doors, drawers and bins	Not secured as specified	103	
	Keys not bagged and secured to handle (when specified)	104	
	Cushioning material missing (when specified)	105	
	Strap not centered on cushioning material		205
	Strap tensioned too tightly causing damage	106	
	Strap tensioned too loosely causing slippage	107	
	Not taped as specified (when required)	108	
Legs	Not removed when detachable	109	
	Hardware missing from package	110	
	Not cushioned and packaged as specified	111	

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the currentversion before use.

TABLE III. End item preservation and packaging defects (cont'd)

Examine	Defect	Category Major	Category Minor
Leveling pads	Not completely screwed in, or not detached and packaged, as specified		206
Line cord	Not looped and tied securely as specified		207
	Voltage and frequency notice missing or not attached (when specified)	112	
Accessories and parts (boxed)	Not individually wrapped with cushioning material	113	
	Not packaged in box specified	114	
	Box not closed as specified	115	

TABLE IV. End item packing defects

Examine	Defect	Category Major	Category Minor
Levels A and B			
Option I			
Refrigerator	Not secured and cushioned within container as specified (see table I and 3.3)	101	
	Finished surfaces not protected from abrasion with material specified	102	
Accessories and parts box	Not securely blocked and braced within refrigerator (when specified)	103	
	Not secured within the shipping container (when specified)	104	
	More than two shipping containers per complete refrigerator when packed separately	105	
Container	Not as specified for weight limits of refrigerators (see 3.2.1.1.1, 3.2.1.1.2, and 3.2.1.1.3)	106	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

TABLE IV. End item packing defects (cont'd)

Examine	Defect	Category Major	Category Minor
Container (cont'd)	Less than minimum clearance specified between refrigerator and inside of container	107	
Waterproof barrier	Waterproof barrier missing (when specified)	108	
	Less than 4 inch lap at joints		201
Bolts and nuts	Less than four bolts holding refrigerator	109	
	Diameter of bolt less than specified	110	
	One or more flat washers missing under heads of machine bolts	111	
	One or more nuts loose on bolts	112	
	One or more lockwashers missing	113	
	Bolts not positioned through container members and blocking as specified	114	
Blocking members	Dimensions less than specified	115	
Edge pad	Not built-up corrugated fiberboard	116	
	Less than length required to cover edge	117	
	Less than 3 inches of pad contacting each adjacent corner surface of refrigerator	118	
	Does not provide minimum 1 inch clearance beyond outermost projection	119	
	Not fitted as required to make tight pack	120	
Tray	Not double faced corrugated	121	
	Less than 4 inches deep	122	
	Missing on top or bottom	123	
Wood frame	Width and thickness less than specified	124	
	Not secured to base as specified	125	
	Missing on top or bottom	126	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

TABLE V. End item marking defects

Examine	Defect	Category Major	Minor
Marking			
Package and shipping container as applicable	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application	101	

4.2.3 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-621, PPP-B-636, PPP-B-640, PPP-C-650, MIL-C-104, and MIL-C-11133, examination of defects in closure and strapping shall be in accordance with the applicable container specification.

4.2.3.1 Examination of shipping containers when PPP-B-576 and PPP-B-601 are required. When shipping containers are required to be in accordance with PPP-B-576 and PPP-B-601, examination of defects in closure and strapping for determination of compliance with the specification, as applicable, and with Drawing No. 5-13-1153 shall be in accordance with the defects listed in table VI. The lot size shall be expressed in shipping containers. The sample unit shall be one shipping container. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE VI. Strapping and closure defects

Examine	Defect	Category Major	Minor
Strapping	Not applied as specified (Drawing 5-13-1153)	101	
	Missing strap	102	
	Loose strap	103	
	Torn strap	104	
	Cut strap	105	
	Not drawn tight	106	
	Weak joint	107	
Staples	Staples less than 4 inches from edge of member		201
	Less than size specified	108	
	Not spaced as specified		202

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:42Z
Check the source to verify that this is the currentversion before use.

4.2.4 End item testing. When applicable, one sample unit shall be selected at random from each lot and tested for compliance with 3.1.1.1.4, 3.1.1.1.4.2, 3.1.1.1.6, and 3.1.1.1.9 in accordance with 4.3. Any test failure shall be cause for rejection of the lot.

4.3 Test methods.

4.3.1 Staining test. The staining test shall be performed as specified in 4.3.1.1, 4.3.1.2, and 4.3.1.3.

4.3.1.1 Conditioning. The tape shall be conditioned for testing for not less than 24 hours at 73°F, ± 2°F, and 50 percent ± 4 percent relative humidity.

4.3.1.2 Specimens. The first three layers of tape shall be removed from the rolls before taking specimens to be tested. An equal number of 10 inch lengths of tape shall be taken as specimens from all sample rolls to be tested.

4.3.1.3 Staining. Specimens of tape shall be applied, adhesive down to a clean porcelain enamel surface panel and rolled down with two passes, once in each longitudinal direction, using a 4-1/2 pound rubber roller having a surface durometer hardness between 70 and 80 and approximately 3-1/2 inches in diameter, at a rate of travel of 10 inches per minute. Panels shall be placed in an oven having a temperature of 150°F ± 2°F for a period of 10 days. Upon removal, panels shall be allowed to cool 2 hours before stripping off the tape. Tape edge transfer may be noticed due to the high temperature. If so, it should be removed with solvent prior to determination for staining. Tape shall be rejected for use if staining is evident.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The packaging and packing of refrigerators and related equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

c. Options required (see 3.1.1).
d. Levels of packaging and packing required (see 3.1 and 3.2).
e. When weather resistant fiberboard is required (see 3.2.2.1.1, 3.2.2.1.2, and 3.2.2.1.3).

6.3 Subject term (key word) listing.

Bins
Cleaning
Doors
Draining
Drawers
Legs
Options

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

(Project 4110-0469)

Review activities:

DLA - GS

User activities:

Army - ME
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-12323G	2. DOCUMENT DATE (YYMMDD) 1992 December 31

3. DOCUMENT TITLE
REFRIGERATORS AND RELATED EQUIPMENT, PACKAGING AND PACKING OF

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil - Downloaded: 20260417T22:42Z
Check the source to verify that this is the current version before use.